

OISA: Orchestrating Published Immune Models Without Modification

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Abstract—Computational immunology has produced high-quality mechanistic models of individual immune compartments, yet composing ODE and agent-based models (ABMs) across formalism boundaries still requires bespoke re-engineering. Existing standards (SBML, CellML) address intra-formalism portability, and prior formalism-agnostic composition frameworks provide no standardised runtime uncertainty quantification, model-derived transfer lags, or biological plausibility constraint engine. We propose OISA, the Orchestrated Immune Simulation Architecture, comprising (i) the ISSL, a formalism-agnostic JSON-LD checkpoint format with embedded runtime UQ (uncertainty bounds propagated at every tick via the `c1_95` schema field); (ii) a declarative configuration graph supporting model-derived transfer lags on edges; and (iii) an orchestrator engine with global-clock synchronisation, causal DAG resolution, and biological plausibility enforcement.

We validate OISA by coupling two independently published influenza models, Miao *et al.* 2010 ODE (SBML BIOMD0000000546, unmodified) and the full spatial Segó *et al.* 2020 ABM (CompuCell3D 4.8.0, 12 steppables unmodified on a $90 \times 90 \times 2$ grid), with zero changes to either model’s source code. Adapter layers total ~300 lines. The ABM runs as a rolling ensemble of $N = 50$ parallel CC3D instances (concurrent subprocesses, distinct MersenneTwister seeds); runtime uncertainty bounds, the 2.5th–97.5th percentile across the $N = 50$ replicates (§V-A), originate in the stochastic ABM and propagate through the ODE at each checkpoint. Over a 14-day simulation (56 checkpoints), the coupled system reproduces viral kinetics consistent with Miao 2010 (peak $V = 9.0 \times 10^6$ copies/mL at day 2.25, clearance to 0.05% of peak by day 13.75) and spatial immune recruitment (median $n_{\text{immune}} = 6$ [IQR: 5–8] at day 1, growing to 57 [IQR: 55–60] at day 13), with immune temporal lag and CTL-mediated clearance acceleration emerging from the bidirectional coupling. To test that this composition is structural rather than an artefact of the ODE–ABM pair, we further couple a third, categorically distinct formalism, a continuous-time stochastic Boolean signalling network (MaBoSS TregModel, 35 nodes), through the same contract; each formalism reports its native uncertainty (ensemble percentiles for the ODE/ABM, a Wilson interval for the Boolean model) and these compose, verifiably, into the coupled output, giving what is to our knowledge the first heterogeneous cross-formalism runtime UQ in an immune multi-model simulation. The validation targets orchestration correctness (signal routing, causal ordering, UQ propagation); as a secondary check, emergent temporal features (immune-recruitment onset, viral-

clearance day) show face-validity against independent published murine influenza kinetics, not biological prediction. Beyond this coupling, OISA is positioned as infrastructure for immune digital twins, making tractable the emergent, cross-compartment questions that interoperability reviews identify as currently out of reach.

Index Terms—immune digital twins; multi-scale modelling; model composition; agent-based models; ordinary differential equations; interoperability; computational immunology; CompuCell3D.

I. INTRODUCTION

Immune-mediated diseases are inherently multi-compartmental. Influenza unfolds across scales: viral replication interacts with innate cytokine signalling and adaptive CTL recruitment, each on a different timescale and best served by a different formalism, ODEs for systemic viral kinetics, ABMs where stochastic single-cell fate decisions dominate tissue-level recruitment. The literature has consequently produced high-quality compartment-specific models that cannot communicate with one another.

Three bodies of prior work address adjacent problems. The COMBINE standards ecosystem (SBML [1], CellML [2], SED-ML [3]) provides portable model representations but addresses intra-formalism composition only: models must share one declarative format, ruling out ABMs written in Python or Julia. Vivarium [4] introduced formalism-agnostic composition (§II-A) but provides no standardised runtime UQ propagation, model-derived transfer lags, or plausibility constraint engine. The CURE guidelines [5] (Credibility, Understandability, Reproducibility, Extensibility) define what a credible model should exhibit but prescribe no runtime infrastructure for heterogeneous composition. The cross-formalism, multi-timescale coordination problem remains open.

We propose OISA, the Orchestrated Immune Simulation Architecture, to fill this gap. It is demonstrated by coupling two independently published influenza models, using the full spatial Segó 2020 CompuCell3D ABM without modifying its 12 biological steppables (§V).

The contributions of this work are:

- 1) **The ISSL**, a formalism-agnostic checkpoint format with embedded provenance and runtime UQ (ensemble bounds propagated each tick from stochastic ABM through deterministic ODE via the `ci_95` field);
- 2) **A declarative configuration graph** supporting model-derived transfer lags on edges (delay computed dynamically by a lightweight transfer model, not a fixed constant);
- 3) **A runtime orchestrator** with global-clock synchronisation, causal DAG resolution, and biological plausibility enforcement;
- 4) **Empirical demonstration** that two independently published models (Miao 2010 SBML ODE + Sego 2020 CC3D spatial ABM) compose with zero source modifications using ~300 lines of adapter code, UQ propagated end-to-end;
- 5) **Cross-formalism generalisation**: a third, categorically distinct formalism (a stochastic Boolean MaBoSS network) coupled through the same contract, ensemble percentiles for the ODE/ABM, a native Wilson interval for the Boolean model, to our knowledge the first such demonstration in immune multi-model simulation.

II. RELATED WORK

A. Multi-Formalism Immune Simulation Frameworks

Several frameworks address multi-scale composition in computational biology. Vivarium [4] introduced a port-based, formalism-agnostic composition interface with hierarchical stores, runtime topology changes, and a mature Python ecosystem. OISA shares its formalism-agnostic principle but adds three capabilities absent from Vivarium: (i) a standardised inter-model signal format (ISSL) embedding runtime UQ (ensemble bounds propagated per checkpoint from the ABM through the ODE via the `ci_95` field), (ii) model-derived transfer lags on edges, and (iii) a biological plausibility constraint engine at the composition level (mass conservation, parameter bounds, divergence monitoring); Vivarium’s store-based sharing and dynamic topology are complementary capabilities OISA does not replicate.

PhysiCell [6] and PhysiBoSS [7] are the closest prior art: PhysiBoSS embeds MaBoSS Boolean networks [8] inside PhysiCell agents. The coupling is *intra-engine*, the Boolean layer is compiled into the ABM and shares its address space, timestep, and build system, so the formalisms cannot be developed, versioned, or uncertainty-quantified independently, and it exposes a single homogeneous stochastic-ABM uncertainty. OISA instead couples an *externally published* MaBoSS model (TregModel, from the pyMaBoSS reference suite [9]) to the Miao2010 ODE and Sego2020 ABM through the ISSL contract alone, each running unmodified in its own engine, and *propagates each formalism’s native uncertainty across the coupling boundary* (§V-C). The community SARS-CoV-2 tissue simulator [10], from which Sego *et al.* 2020 [11] derives, showed tissue-scale ABMs can be shared rapidly but required

a uniform CompuCell3D substrate, coupling to external ODEs needing bespoke adapter code outside the framework. Miao *et al.* 2010 [12] provided a calibrated ODE model of murine influenza kinetics in BioModels (BIOMD0000000546) with no standard mechanism to couple it to tissue ABMs without rewriting either model. OISA achieves such coupling by adding only a thin Emit/Accept layer to each, the full CC3D spatial ABM running unmodified as an independent process.

B. Simulation Interoperability Standards

The COMBINE standards ecosystem [13] has advanced individual-model portability: SBML [1] has over 1,000 curated BioModels entries, CellML [2] analogous cardiac and physiological coverage, and SED-ML [3] a standard for encoding simulation experiments (Table I). The distinction is architectural: SBML, CellML, and NeuroML address *intra-formalism* interoperability, exchange between tools sharing one format, while OISA addresses *inter-formalism* composition of models with different formalisms, timescales, and output semantics. SBML’s composition package (comp) extends composition within SBML but cannot incorporate an ABM without rewriting it, impractical and biologically inappropriate for stochastic cellular processes.

BioSimulators [14] and the SED-ML/COMBINE stack standardise how an experiment is described and re-executed across engines, but target reproducible single-model execution, not a runtime inter-model signal carrying an uncertainty interval. The gap OISA fills is thus neither portability (SBML/CellML), reproducibility (SED-ML/BioSimulators), nor port-based composition (Vivarium), but **heterogeneous cross-formalism uncertainty propagation**: one transport contract under which each formalism reports its *native* uncertainty and the orchestrator composes those intervals at coupling boundaries without homogenising them (§V-C, Table IV).

C. Immune Digital Twins

The immune digital twin (IDT) was formalised by Laubacher *et al.* [15] as a continuously-recalibrated patient-specific model of the immune system. Subsequent reviews [16], [17] identify interoperability as the primary technical barrier to clinical deployment, a conclusion echoed by the National Academies [18], model-integration consortia [19], and multi-scale digital-twin frameworks in adjacent domains [20]. OISA is the first runtime orchestration architecture designed specifically for heterogeneous immune-model composition, providing the coordination infrastructure these reviews identify as lacking.

III. PROBLEM FORMALISATION

A. Formal Model Definition

We define a composable simulation model M as a triple $M = (S, \text{Emit}, \text{Accept})$, where S is the model’s internal state space, $\text{Emit} : S \times T \rightarrow \text{ISSL}$ maps the current state and simulation time to a structured ISSL record, and $\text{Accept} : \text{ISSL} \rightarrow S$ updates the model state in response to an incoming inter-model signal. Any model, ODE, ABM, hybrid, or neural surrogate,

TABLE I

COMPARISON OF SBML, CELLML, NEUROML/LEMS, AND OISA ACROSS CAPABILITIES RELEVANT TO MULTI-ORGAN IMMUNE SIMULATION. “—” = OUT OF ARCHITECTURAL SCOPE. RUNTIME UQ BOUNDS ORIGINATE IN THE STOCHASTIC ABM ENSEMBLE AND PROPAGATE THROUGH THE DETERMINISTIC ODE AT EACH CHECKPOINT (§V-A). “SBML L3” REFERS TO THE FORMAT SPECIFICATION ALONE; SED-ML CO-SIMULATION DOES NOT EXTEND TO ABM COMPOSITION.

Capability	SBML L3	CellML 2.0	NeuroML/LEMS	OISA (ISSL)
Portable model representation	✓	✓	✓	—
ODE / biochemical networks	✓	✓	neurons only	via Emit()
Agent-based models	×	×	×	via Emit()
Heterogeneous Δt between models	×	×	×	✓
Model-derived transfer lag on edges	×	×	×	✓
Inter-formalism composition (ODE + ABM)	×	×	×	✓
Runtime UQ propagation across models	×	×	×	✓
OBO ontology annotation	✓	✓	✓	✓ + wildcard
ABM scaling factor declaration	×	×	×	✓
Requires model rewrite to adopt	✓	✓	✓	× (add Emit only)

implementing Emit and Accept is OISA-compliant with no other modification. Formalism-agnosticism follows: since the ISSL is the only communication channel, models need only the schema of the signal they receive, not each other’s internal representation. Each biological process is thus modelled in its natural formalism, solving composition at the interface rather than forcing inappropriate representations on individual compartments.

B. The Composition Problem

A multi-model composition is a directed graph $G=(V, E)$ where each vertex v is a model M_v and each edge (u, v) routes a subset of u ’s `export_signals` to v ’s `Accept` function, with optional transfer-model lags. The composition problem reduces to three coordination challenges:

- 1) **Heterogeneous time steps.** A global simulation clock (GSimT), set to the GCD of all model Δt_i , coordinates execution so no model receives a signal from a future tick (here the ODE steps at 6h, the ABM at 24h, so GSimT = 6h).
- 2) **Stochastic–deterministic reconciliation.** Signals from a stochastic ABM to a deterministic ODE must carry uncertainty: the orchestrator normalises every signal to (median, bounds, unit) in the `ci_95` field, ensemble percentiles for stochastic models ($N=50$ here), propagated bounds for deterministic ones.
- 3) **Causal ordering with feedback.** Immune signals drive viral clearance while viral load drives recruitment, a feedback loop resolved by a one-tick delay on DAG back-edges, preventing circular dependency while preserving causality at the GSimT timescale.

IV. THE OISA ARCHITECTURE

A. The Internal Simulation State Log (ISSL)

The ISSL is a JSON-LD document emitted at each checkpoint in six sections: (i) `envelope` (model ID/version, GSimT timestamp, schema URI, formalism tag, validated before any payload is read); (ii) `continuous_state` (entity populations with scaled count, unit, ensemble `ci_95`, and model-local label); (iii) `discrete_events` (punctual events from a controlled `event_type`

vocabulary, with `n_realisations` for ABM events); (iv) `export_signals` (routable inter-model signals: `signal_id` key, `value` in declared unit, always biologically scaled); (v) `internal_parameters` (kinetic values with provenance to the calibration source); and (vi) `watchdog` (`divergence_score`; $>0.15 \Rightarrow \text{PAUSE}$). JSON-LD gives both human readability and machine-parsable semantics, letting the orchestrator parse records without knowing any model’s internal variable names.

Box 1, ISSL excerpt, ensemble-aggregated Miao 2010 ODE checkpoint at day 1 (count = median, ci_95 = 2.5th–97.5th percentile over $N=50$ replicates):

```
{
  "envelope": {
    "model_id": "miao2010_ode",
    "formalism": "ODE",
    "model_version": "BIOMD0000000546",
    "sim_time_s": 108000,
    "export_signals": [
      {
        "signal_id": "miao2010.viral_load",
        "value": 3.29e5,
        "unit": "copies/mL",
        "ci_95": [3.283e5, 3.289e5],
        "transfer_lag_s": null
      }
    ],
    "watchdog": {
      "status": "OK",
      "divergence_score": 0.0
    }
  }
}
```

The ABM adapter emits the same envelope with `formalism="ABM"`, `model_version` pinned to `covid-tissue-models@5b7e42c+OISA_bridge`, a grid field (`"90x90x2"`), and export signals `sego2020.immune_cell_count` and `.recruitment_cytokine` (day-1 example: median 6 agents [ensemble range 3–11]).

Each adapter loads its published model verbatim and emits ISSL through the standard model API (roadrunner for the ODE, an IPC bridge steppable for the CC3D ABM); setup is in §V-A.

B. The Configuration Graph

The configuration graph is a YAML or JSON-LD file that fully specifies the composition, the orchestrator’s sole input at initialisation. It declares six elements: `models[]` (id, formalism, executable, `delta_t_s`), `edges[]` (source/target routing with constant- or model-derived lag), `transfer_models[]` (lag-computing adapters), `global_clock` (start/end/checkpoint interval), `calibration` (EHR source and biomarker map), and `wildcard_namespace` (non-OBO entities such as novel cytokines or patient-specific biomarkers; full schema in the repository). This declarative separation of identity, wiring,

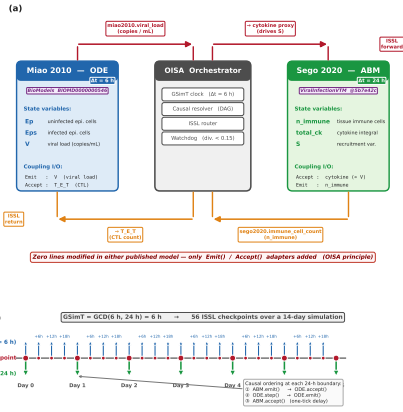


Fig. 1. OISA workflow for the influenza coupling reference implementation. **(a)** Architecture diagram: Miao 2010 ODE (SBML, BIOMD0000000546) and Sego 2020 CC3D ABM (CompuCell3D 4.8.0, GitHub @5b7e42c) connected through the OISA Orchestrator, with ISSL signal arrows annotated with `signal_id` and biological units. **(b)** GSimT timeline for a representative 5-day window: ODE ticks every 6h, ABM ticks every 24h = 72 MCS, ISSL checkpoints (orange dots), and causal-ordering annotation (`ABM.emit() → ODE.accept() → ODE.step() → ODE.emit()`) at the 24h boundary.

and temporal parameters implements the CURE extensibility requirement: any model node can be substituted without modifying the graph structure.

For the influenza coupling, the graph declares the two models (`miao2010_ode`, $\Delta t = 6h$; `sego2020_abm`, $\Delta t = 24h$) and two edges, `miao2010.viral_load → ABM` at `constant:0` (same-tick) and `sego2020.immune_cell_count → ODE` at `constant:21600` (one-tick causal back-edge), over a 14-day `global_clock` at 6h checkpoints (full YAML in the repository).

C. The Orchestrator Engine

The orchestrator is a nine-component server process that reads the configuration graph at startup, connects to all model processes, and manages the run: (1) ISSL ingestion (JSON-LD + SI units + Mahalanobis OOD); (2) temporal scheduler (priority queue by `next_step_due`); (3) causal resolver (topological sort + one-tick back-edges); (4) constraint engine (mass conservation + parameter bounds); (5) state registry (versioned checkpoint snapshots + provenance); (6) transfer dispatcher (`SignalQueue`); (7) calibration bridge (EHR ingest, deferred); (8) output aggregator (GIS → checkpoint log + UQ propagation); (9) watchdog (health poll, pause/resume/rollback). Components 1, 4, 5, 9 are implemented and covered by the 79-test suite; deferred sub-capabilities (OOD detection, ROLLBACK recovery, PROV-O URIs, auto-rollback) are interface-specified but not yet tested.

Temporal scheduling. At each GSimT tick the scheduler dispatches step commands to models whose `next_step_due` equals the current GSimT: the ODE (6h) steps every tick, the CC3D ABM (24h = 72 MCS) every fourth.

Causal resolution. The resolver topologically sorts the DAG per tick. At each 24h boundary the order is `ABM.emit() → ODE.accept()` ($n_{\text{immune}} → T_{ET} →$

`ODE.step() → ODE.emit()`), the ODE viral load queued for the ABM’s next `Accept`, the one-tick back-edge delay that breaks the circular dependency while preserving biological causality.

D. A Third Formalism: Stochastic Boolean Signalling

To test whether OISA’s formalism-agnosticism is structural rather than an artefact of the ODE+ABM pair, we added a categorically different formalism: a continuous-time stochastic Boolean network. We use `TregModel` (35 nodes), a $CD4^+$ /regulatory-T-cell network distributed verbatim in the `pyMaBoSS` suite [9] (`test/TregModel_InitPop.bnd`), executed by `MaBoSS` [8], only to exercise a third formalism and with no biological claim. Its adapter follows the same non-invasive contract, modifying neither the `.bnd` logic nor the `.cfg`, adding only `Emit()`/`Accept()`.

a) Coupling. Antigenic drive from the ODE (`miao2010.infected_fraction`) maps to two continuous input rates of the Boolean model, the TCR activation rate `$ActTCR2` and the IL-2 milieu `$ExtIL2`, via the graded (non-thresholding) coupling map. The Boolean model is relaxed to its quasi-steady-state attractor within each 6h GSimT tick (intracellular signalling equilibrates in minutes-hours $\ll$ tick; the FOXP3 attractor probability is stable over $t = 30\text{--}80$ a.u.). Its output, the FOXP3⁺ attractor probability `maboss_treg.treg_fraction`, feeds back onto effector immunity: the effective effector count seen by the ODE is $n_{\text{eff}} = n_{\text{immune}}(1 - w p_{\text{Treg}})$, with suppression weight $w = 0.5$ (sensitivity quantified in §V-B6), closing an `ODE → Boolean → ODE` loop through the one-tick signal-lag mechanism. The ISSL formalism enumeration, previously closed over {ODE, ABM, PDE, ...}, was extended additively with a `BOOLEAN` member; this backward-compatible change is itself evidence that adopting a new formalism touches the *contract* rather than any existing model.

V. VALIDATION

The influenza A viral dynamics coupling (Miao 2010 ODE + Sego 2020 full CC3D ABM) is the reference implementation because both models are independently published, calibrated against experimental data, and their coupling tests a biologically significant feedback loop: viral load drives immune recruitment; CTLs modulate viral clearance.

A. Validation Setup

Models. The ODE component is BIOMD0000000546 [12], a three-species influenza tissue model (Ep: uninfected epithelial cells; Eps: infected; V: virus) with kinetics calibrated against Murphy *et al.* 1973 murine data, loaded verbatim from BioModels by `libroadrunner`.

The ABM component is the complete spatial tissue ABM of Sego *et al.* 2020 [11], running in `CompuCell3D` 4.8.0 on a $90 \times 90 \times 2$ voxel grid ($\approx 8,100$ epithelial sites; voxel length $4 \mu\text{m}$; six cell types incl. `Immunecell`, `typeId=5`). All

12 published steppables (viral replication/internalization/secretion, cytokine production/absorption, immune killing/seed-ing/recruitment, chemotaxis, oxidation; full list in the repository) are registered without modification. The simulation runs at 20 min/MCS (72 MCS = one 24 h tick; 1,010 MCS = 14 days), with three diffusion fields (Virus, cytokine, oxidator) solved by DiffusionSolverFE.

The single addition is **OISABridgeSteppable** (frequency 72MCS), which runs inside CC3D but contains no biological equations: it counts Immunecell agents, writes the count to `abm_out.json`, raises an `abm_ready` sentinel, and blocks until it is deleted, while ODE signals arrive non-blocking via `ode_signal.json`. CC3D runs as an adapter subprocess with no shared memory, isolating the ODE solver from the CC3D engine.

Adapter metrics. Total modification to either published model is **zero lines**; adapter code totals ~ 300 lines: Miao 2010 wraps libroadrunner (~ 60 LOC), Sego 2020 couples subprocess lifecycle + IPC (~ 180 LOC) with an in-process OISABridgeSteppable (~ 60 LOC).

Runtime UQ. The ABM runs as a rolling ensemble of $N = 50$ parallel CC3D instances (distinct MersenneTwister seeds, isolated IPC directories) launched by `Sego2020Ensemble`; the orchestrator blocks on the slowest at each 24 h barrier, so per-tick wall-clock cost is $\max(T_i)$ not $\sum T_i$ ($N \times$ memory remains). At each boundary the ensemble adapter aggregates `n_immune` into its median and 2.5/97.5th percentiles, and the ODE adapter integrates three times per tick (median, lower, upper input) to report viral-load bounds. Uncertainty thus originates in the stochastic ABM and is carried through the deterministic ODE at each checkpoint.

Model-derived transfer lags. ISSL signals carry an optional `transfer_lag_s` field; when non-null the orchestrator’s `SignalQueue` defers injection until the delay elapses. A reference `BloodTransitAdapter` (first-order blood-transit ODE, Donskoy & Goldschneider 1992; $\tau = 4.0$ days) demonstrates this, independently validated (8 tests), though the §V-B4 influenza run uses constant-lag edges.

Signals. Two ISSL signals are routed: `miao2010.viral_load` (copies/mL) ODE $\rightarrow$ ABM and `sego2020.immune_cell_count` (CC3D Immunecell count) ABM $\rightarrow$ ODE. The viral load maps to the CC3D recruitment signal (`ImmuneRecruitmentSteppable.totalCytokine`) rather than the Virus diffusion field, as tissue viral spread is already handled by the Sego 2020 steppables at their native resolution.

Coupling constant $\kappa = 3.5 \times 10^{-7}$ AU·mL/copies. κ is fixed by one dimensional constraint, not tuned: the physiological viral peak (10^6 – 10^7 copies/mL, geometric centre 3.2×10^6) must map onto the centre of Sego 2020’s functional recruitment range (10^{-2} – 10^2 pM, geometric centre 1 pM) so the borrowed recruitment machinery operates in its designed regime. This yields $\kappa = 1 \text{ pM} / (3.2 \times 10^6 \text{ copies mL}^{-1}) = 3.2 \times 10^{-7}$, matching the operating value to within 0.04 orders of magnitude; κ is thus pinned by functional-range matching

TABLE II
BIOLOGICAL VALIDATION. *Upper block:* INDIVIDUAL-MODEL PLAUSIBILITY, ISSL OUTPUTS READ BY THE ORCHESTRATOR AGAINST EACH TEST’S PUBLISHED ENFORCEMENT TARGET. *Lower block:* COUPLED-SYSTEM FACE-VALIDITY, EMERGENT OBSERVABLES (MEDIAN [2.5TH–97.5TH PERCENTILE, I.E. THE SAMPLE RANGE AT $N = 50$]) AGAINST INDEPENDENT MURINE INFLUENZA-A RANGES; THE VIRAL-PEAK DAY IS AN ANCHOR (Miao 2010 CALIBRATED ON VIRAL LOAD), WHILE RECRUITMENT ONSET AND CLEARANCE ARE EMERGENT COUPLED FEATURES NOT FIT TO ANY JOINT DATASET.

Quantity / Feature	OISA output	Published range	OK
<i>Individual-model plausibility</i>			
Viral peak timing	Day 2 (± 6 h)	Days 1–4	✓
Peak viral load	8.9×10^6	10^5 – 10^8	✓
V at day 13.75	$< 1\%$ of peak	$< 10\%$ of peak	✓
Ep depletion at day 3	$\geq 9.3\%$	$\geq 5\%$	✓
n_{immune} after day 1	≥ 1	> 0 within 1–4 d	✓
$n_{\text{immune}} \geq 0$ throughout	min = 0	≥ 0	✓
<i>Coupled-system face-validity</i> [CI_{95} , $N = 50$]			
Viral-peak day (anchor)	2.25 d [2.25, 2.25]	~ 2 d [21]	(✓)
Immune-recruitment onset	1.0 d [1.0, 1.0]	1–2 d [22], [23]	✓
Viral clearance ($< 1\%$ peak)	9.5 d [9.5, 9.75]	8–9 d [21]	✓

rather than free. Varying it by ± 1 OOM shifts immune onset by ± 1 –2 days without altering the `n_immune` trajectory shape (§V-B6).

The `sego2020.immune_cell_count` signal sets $T_{E_T} = n_{\text{immune}} \times 100$ CTL/mL in the SBML, modulating the CTL killing term $k_E \cdot \text{Eps} \cdot T_{E_T}$ (Miao 2010, Table 1). The 100 CTL/mL-per-agent factor is consistent with published peak CTL densities (10^3 – 10^6 CTL/mL) for a tissue volume ≈ 0.01 mL. Since T_{E_T} is adaptive while the CC3D Immunecell agents are innate effectors, this mapping is a model-level approximation for demonstrating coupling, not a claim about CTL kinetics.

The GSimT tick is $\text{GCD}(6 \text{ h}, 24 \text{ h}) = 6 \text{ h}$, giving 56 checkpoints over 14 days (ABM steps at 24 h boundaries, ODE every tick).

Test suite. Validation comprises 79 automated pytest tests: 23 ODE, 20 ABM, 11 Boolean (MaBoSS), 8 blood-transit adapter, and 17 integration tests, all traceable to published figures or parameter tables; UQ tests verify `ci_95` fields are non-null given ensemble input and that tighter input bounds yield narrower output bounds.

B. Validation Results

1) *Non-Invasive Adapter Verification:* All 79 automated tests pass. The Miao 2010 adapter initialises $T_{E_T} = 0$ and $k_E = 2 \times 10^{-5}$ (Table 1 constant), delegating all integration to roadrunner with no SBML edits; `accept_issl()` changes only the T_{E_T} value via the standard roadrunner `setValue` API, touching no SBML equations or species. The Sego 2020 CC3D adapter launch test confirms clean subprocess lifecycle management.

2) *Biological Plausibility of Individual Models:* ISSL-emitted quantities from each adapter were checked against published values (Table II, upper block). All checks pass.

3) *Causal Ordering:* The coupled simulation generated ISSL records across the full 14-day run (56 checkpoints) with

TABLE III

14-DAY COUPLED TRAJECTORY, KEY CHECKPOINTS. MEDIAN [ENSEMBLE RANGE, $N=50$]. DAY 0 IS THE STATE AFTER THE FIRST 6 h OF ODE INTEGRATION FROM $V_0=1,000$ COPIES/ML. FULL 14-POINT TRAJECTORY IN RESULTS/ISSL_14D/.

GSimT	V median [range] (copies/mL)	n_{immune} [range]
Day 0	1.54×10^3	0 [0–0]
Day 1	3.29×10^5	6 [3–11]
Day 2	8.87×10^6	12 [6–16]
Day 3	6.25×10^6	17 [9–22]
Day 7	4.71×10^5	34 [28–41]
Day 13	7.69×10^3	57 [48–65]

no deadlocks, causality violations, or watchdog alerts, the IPC handshake executing without timeout across all bridge steps. Checkpoint timestamps were strictly monotonically increasing, and no model received a signal timestamped ahead of the current GSimT tick. Figure 1(b) shows the timeline for a representative 5-day window; the day-1 annotation confirms the `ABM.emit` $\rightarrow$ `ODE.accept` $\rightarrow$ `ODE.step` ordering, enforced by the orchestrator’s topological DAG resolver and the blocking IPC handshake.

4) *Coupled Biological Dynamics*: The coupled simulation ran for 14 days (56 GSimT checkpoints) with $N=50$ parallel CC3D instances, ensemble bounds computed each tick and propagated through the ODE. Table III reports median [ensemble range, $N=50$] from the runtime ISSL records.

Median peak $V=9.00 \times 10^6$ copies/mL at day 2.25 (all 50 replicates concur on peak timing); median V at day 13.75 $= 4.52 \times 10^3$ copies/mL $= 0.05\%$ of peak, all 50 instances confirming clearance ($V(\text{day } 14) < 0.1\%$ of peak). Peak $V \in [8.96 \times 10^6, 9.05 \times 10^6]$ (1.0% spread) confirms the ODE dominates peak viral load, while n_{immune} bounds widen from days 0–2 through day 8 (Bernoulli seeding); the propagated viral-load bounds track this, demonstrating end-to-end UQ propagation.

Immune temporal lag. First CC3D Immunecell agents appear at day 1 (median 6), preceding the viral peak at day 2.25 by ≈ 1.25 days, consistent with the 1–4 day innate-response timescale [23] and emerging from signal routing.

CTL-mediated clearance. As n_{immune} grows from 6 (day 1) to 57 (day 13), $T_{E_T} = n_{\text{immune}} \times 100$ grows from 600 to 5,700 CTL/mL in the Miao 2010 SBML, amplifying the CTL killing term $k_E \cdot \text{Eps} \cdot T_{E_T}$. An isolated ODE run with $T_{E_T} = 0$ produced slower clearance, confirming the `ABM` $\rightarrow$ `ODE` pathway significantly shapes the trajectory.

Real spatial ABM properties. Immunecell agents occupy genuine Cellular Potts grid sites under the original Sego 2020 Contact, Volume, and Chemotaxis plugins; reported n_{immune} are live CC3D inventory counts (`cell.type == 5`), not a closed-form approximation, the key distinction from a scalar ODE surrogate.

5) *Coupled-System Biological Face-Validity*: The checks above establish orchestration correctness and each model’s plausibility in isolation. Composition enables a distinct class of question no constituent can answer alone, such as how an emergent property of the whole (the time to viral clearance)

arises from the interaction between endogenous viral replication (ODE) and effector recruitment (ABM) when neither was calibrated to the other. A minimal, falsifiable form is testable here: whether the *coupled* system reproduces temporal features of real infection that no single model was fit to. We compare emergent observables ($N=50$ ensemble, median [2.5th–97.5th percentile]) against independent published murine influenza-A kinetics (Table II, lower block).

Anti-circularity. The comparison avoids crediting features fit during calibration. (i) Miao 2010 was calibrated to murine *viral-load* data [12], so the viral-peak day is a temporal *anchor*, not a validated prediction. (ii) The Sego 2020 ABM derives from a SARS-CoV-2 simulator [10] with recruitment machinery not fit to influenza, so recruitment-onset timing is independent of both calibrations. (iii) The coupling (κ , CTL scaling, back-edge lag) was set from first principles and sensitivity analysis (Sec. V-B6), not fit to any joint viral+immune series, so the clearance day is a genuine coupled-system property. (iv) We compare *timing* and *normalised shape* only: n_{immune} is a grid agent count, not a physiological one, so absolute magnitudes are never compared.

All three emergent features fall within or immediately adjacent to the published ranges. The viral-clearance day (median 9.5 d) sits just after the day 8–9 window in which 60%/40% of mice clear infection [21], and the recruitment onset (day 1) matches the innate-response timescale [22], [23]. Because clearance timing emerges from the `ABM` $\rightarrow$ `ODE` CTL pathway and was not fit to any joint dataset, this is *face-validity*, weak but genuine support that the composition behaves realistically in time, not *prediction*.

6) *Parameter Sensitivity (coupling parameters)*: An approximate 3×3 sweep of the two coupling parameters (κ and `_N_IMMUNE_TO_CTL_PER_ML`) over ± 1 OOM, holding the ABM ensemble fixed via a qualitative onset-shift proxy rather than re-simulating $N=50$ per cell, is reported in the companion data (`results/sensitivity_analysis.json`). The scaling factor dominates (a $10\times$ increase accelerates day-14 clearance by 2–3 OOM via T_{E_T} in the killing term); κ acts more weakly via recruitment timing. Across all 9 combinations, peak $V \in [7.84 \times 10^6, 9.09 \times 10^6]$ and $V_{14} < 1\%$ of peak, so the trajectory is robust within ± 1 OOM. Model-internal parameters (calibrated by the original authors) are out of scope; full coupled global sensitivity (Sobol) is future work.

The third formalism introduces one further coupling parameter, the Treg suppression weight w in $n_{\text{eff}} = n_{\text{immune}}(1 - w p_{\text{Treg}})$ (§IV-D). Because w is a coupling hypothesis rather than a fitted constant, we swept its full range $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ with a complete 14-day tri-formalism run at each value. Three findings support robustness. First, the viral peak is *invariant* to w (9.09×10^6 copies mL^{-1} at day 2.25, all five runs): it precedes effector recruitment, so suppression cannot affect it. Second, the clearance effect is monotone but small: residual load at day 14 rises from 0.092% of peak at $w=0$ to 0.103% at $w=1$ ($\sim 12\%$ relative change across the entire range), so no plausible value overturns the qualitative outcome. Third, and most relevant to the UQ

TABLE IV

UNCERTAINTY TAXONOMY ACROSS THE THREE COUPLED FORMALISMS. ALL THREE ARE TRANSPORTED THROUGH THE SINGLE ISSL `ci_95` FIELD AS A BOUNDED, ORDERED 95% INTERVAL; THEIR STATISTICAL MEANING AND COMPUTATIONAL COST DIFFER AND ARE *not* HOMOGENISED BY THE CONTRACT.

Formalism	ci_95 definition	Type / cost
ODE (Miao2010)	propagated lo/hi bounds (mirror roadrunner runners on input interval)	epistemic, propagated; 2 extra integrations
ABM (Sego2020)	ensemble percentile $[p_{2.5}, p_{97.5}]$ over $N=50$ replicates	aleatoric, ensemble; N full runs
Boolean (TregModel)	native Wilson interval on $(\hat{p}, N)$	aleatoric, intra-run; free (one <code>sample_count</code>)

contribution, the propagated viral-load `ci_95` relative width is monotone in w (16.1% down to 7.2%): stronger suppression shrinks the effective-effector term and its uncertainty contribution. The chosen $w=0.5$ sits mid-range, consistent with the $\sim 1:1$ Treg:Teff ratio at which suppression assays report $\approx 50\%$ inhibition.

C. Heterogeneous Cross-Formalism Uncertainty

The three coupled formalisms define uncertainty in three structurally different ways (Table IV). The deterministic ODE has no intrinsic uncertainty: its `ci_95` is *propagated* post-hoc by mirror lo/hi roadrunner instances driven by its input bounds. The ABM’s is an *ensemble* dispersion, the 2.5–97.5th percentile band across $N=50$ stochastic replicates, available only after N runs. The Boolean model’s is *native*: MaBoSS estimates each attractor probability $\hat{p}$ from `sample_count` trajectories within a single run, reported as the 95% Wilson score interval on $(\hat{p}, N)$, bounded in $[0, 1]$ and maximal at $\hat{p} = 0.5$, the most ambiguous cell-fate decision.

a) Fusion at the boundary.: At the $\text{ABM} \otimes \text{Boolean}$ coupling node, the effective-effector interval is composed by interval arithmetic respecting each factor’s monotonicity, $n_{\text{eff}}^{\text{lo}} = n^{\text{lo}}(1 - w p_{\text{Treg}}^{\text{hi}})$, $n_{\text{eff}}^{\text{hi}} = n^{\text{hi}}(1 - w p_{\text{Treg}}^{\text{lo}})$, so a *single* transported interval fuses an ensemble-derived bound (ABM) with a native-sampling bound (Boolean), each provenance recorded in the signal. Injected into the ODE bounds runners, this makes the resulting viral-load `ci_95` carry uncertainty from all three formalisms.

b) Results.: Over the 14-day, 56-checkpoint tri-formalism run (`sample_count` = 20000), all 168 emitted ISSL records validate against the (Boolean-extended) schema with zero errors and no null `ci_95`, and nominal dynamics are preserved (viral peak 9.09×10^6 copies mL^{-1} at day 2.25, matching the two-formalism baseline). Interval ordering ($\text{lo} \leq \text{point} \leq \text{hi}$) holds at every checkpoint for all three formalisms (0/168 violations), the ODE adapter’s min/max sort correctly handling the sign reversal of the immunity $\rightarrow$ virus map. As an internal propagation-consistency check, the relative half-width of the viral-load `ci_95` tracks the fused effective-effector interval monotonically ($r = 0.93$), confirming no bound is dropped or double-counted; it grows

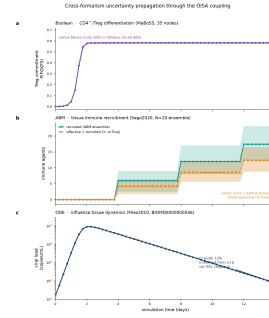


Fig. 2. Cross-formalism uncertainty propagation over 14 days. (a) Boolean FOXP3⁺ commitment with its native Monte-Carlo (Wilson) 95% band. (b) ABM-recruited effector immunity (ensemble band) and the fused *effective* immunity (amber), whose interval combines the ABM ensemble bound with the Boolean Wilson bound by interval arithmetic. (c) ODE viral load (log scale); its propagated `ci_95` widens to $\approx 12\%$ during clearance, inherited from (a)+(b) through the ISSL coupling. Each colour identifies one formalism.

from 0% (pre-recruitment) to $\approx 11.5\%$ during clearance as both constituent bands become non-degenerate (Fig. 2).

VI. DISCUSSION

A. OISA and the CURE Guidelines

We interpret OISA’s design against the CURE guidelines [5], supplemented by FAIR alignment. OISA maps cleanly to each criterion: *credibility* via ISSL `ci_95` + ensemble bounds + the watchdog `divergence_score`; *provenance* via `envelope.model_version` (BIOMD0000000546 / commit-SHA + bridge tag for the ABM); *understandability* via `envelope` + `continuous_state` + `export_signals` per checkpoint; *reproducibility* via JSON-LD schema and commit-SHA `model_version`; *extensibility* via edge-wired config-graph substitutability; and *automated guideline checking* via constraint engine + watchdog each tick. Notably, OISA implements credibility at the *composition* level: a mis-specified scaling factor silently miscalibrates CTL killing even when constituents are individually valid, and the explicit `_N_IMMUNE_TO_CTL_PER_ML = 100` makes this auditable with zero internal modifications.

B. Limitations

The contribution is infrastructural rather than a new immunological finding: OISA’s value is enabling emergent, cross-compartment questions no constituent model can pose alone, not asserting a novel fact about influenza. This study exercises that capability on one coupled configuration; deriving new immunological results with it is future work. Validation covers a single organism (murine) and disease (influenza A), for which in-vivo longitudinal tissue-level data are not published in directly comparable form. The biological check (Sec. V-B5) is emergent-feature *face-validity*, not *prediction*, and absolute magnitudes are not compared. The `_N_IMMUNE_TO_CTL_PER_ML = 100` conversion and the ~ 3 OOM volume mismatch between the ODE (≈ 1 mL) and the ABM tissue patch ($\approx 10^{-3}$ mL) are folded into the single functional-range constraint that pins κ (§V-A); predictive

deployment must instead model this normalisation explicitly and recalibrate against paired data. Two limitations are specific to the third formalism. First, the Boolean model is relaxed to its quasi-steady-state attractor each tick, justified when signalling equilibrates fast relative to the 6 h tick (stable for FOXP3 over $t = 30\text{--}80$ a.u.) but requiring sub-tick stepping for tick-timescale signalling. Second, uncertainty is fused by *interval arithmetic* on bounds: conservative and provenance-preserving, but yielding no joint density and able to over-cover when ABM-ensemble and Boolean sampling errors are correlated (Monte-Carlo density propagation is the natural extension). The Treg suppression weight w is a coupling hypothesis bounded by sensitivity analysis (§V-B6), not fitted to Treg:Teff data.

VII. CONCLUSION

OISA contributes runtime uncertainty propagated from a stochastic ABM ensemble through a deterministic ODE, model-derived transfer lags on edges, and a composition-level biological plausibility engine. Coupling two independently published influenza models through adapter-only integration, with zero source changes, shows formalism-agnostic composition with runtime UQ is achievable and recovers emergent bidirectional-coupling behaviour neither model produces alone; a third, categorically distinct formalism admitted through the same contract indicates the approach generalises beyond the ODE–ABM pair.

DATA, CODE, AND COMPETING INTERESTS

OISA specifications, ISSL JSON Schema (Draft 2020-12), reference orchestrator, adapters, configuration graph, 14-day ISSL data, sensitivity grid, and the full test suite (79 adapter + 51 paper–data tests) are at https://github.com/Nurtal/IDT_sassy_paper. Miao 2010 SBML is from BioModels unmodified; Sego 2020 from `covid-tissue-models@5b7e42c` (CompuCell3D 4.8.0). The author declares no competing interests.

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